PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Stitch - Google Stich

Version: 2.0

1. Project Overview

- **Problem Statement:** [e.g. Design was rigid and specialized skill. Non-designers, especially entrepreneurs/founder/project managers, often find it difficult to visualize their innovative ideas without the specialized software & skills which are expensive in nature. Similarly, professional designers find building layouts from scratch tedious.]
- **Target Domain:** [e.g. Enabling rapid-fast UI/UX for prototypes and early-stage of the solution-based product development]
- **Proposed Solution Summary:** [e.g. Stitch removes the friction of hassle by enabling anyone to prompt an interface in simple English language and receive a fully customized, high-fidelity design & mockups instantly. Enabling the user's role shift from builder to Curator.]

2. Background & Business Objective

- **Background of the Problem:** [e.g. For many years, the design of application followed a manual basis linear pathway. A project manager drafted a document, a graphic designer designed using static shapes in tools i.e. Figma. After that, a developer writes code according to the design. This process requires extensive labor work, specialized software, and the skills limited to certain groups blocked others the opportunity to visualize.]
- **Importance of Solving This Issue:** [e.g. Accelerating the time to prepare a mockup from days to seconds, the industry can test the interface ideas and discard accordingly at a very fast rate. Breaking the technical skill barrier, anyone can visualize their intent without need for years of training.]

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- **Strategic Fit / Impact:** [e.g. It aligns with Google's AI-Native Development Lifecycle Ecosystem. Google's Flagship Gemini model is integrated and export facility to AI Studio, Cloud & Firebase to facilitate developer's entire workspace as google.]

3. Product Purpose

3.1. Main Goal of the System: [e.g. To provide an AI-enabled partner in design.]

3.2. Intended Users (Target Audience):

- Product Manager
- Founder of Products (Early Stage)
- Entrepreneurs
- UI/UX Designer
- Frontend Developers
- Low Code Builders

4. System Functionalities

4.1. Description: [e.g.The system operates as a high-speed generative engine for design purposes through natural language. Using Prompt-to-UI model, intelligence captures required relationships, user flows, multi-screen layout and many more designs accordingly.]

4.2. Key Functionalities:

- **Vibe Designing: [e.g.**Translate human-language descriptive instructions into high-fidelity designs]
- **Conversational Editing:** [e.g.Facilitates continuous modification of the existing design by speaking to canvas itself/ texting.]
- **Versioning & Infinite Canvas:** [e.g. AI Agent Enabled version tracking with dozens of design variation generations is provided.]

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- **Export- Component Based:** [e.g. Enables conversion of the visual into code snippets also editable Figma tool layers.]

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Do state the LLM used (Z AI GLM) along with justification why and what capabilities made it the right choice for the product's core functionalities.

4.3.2. **Prompting Strategy:** Describe your team's approach [e.g. zero-shot, few shot or multi-step agentic prompting]. So, explain the reason why the strategy chosen does fit your use case.

4.3.3. **Context & Input Handling**: Describe how the system that your team developed handles the unstructured or inputs that are oversized. So, what's the maximum input size your product does accept before it actually does truncation or chunking or rejection? So, what's going to happen to those inputs that exceed the limit?

4.3.4. **Fallback & Failure Behavior:** What does the developed product do when the model returns off-topic or unusable and most importantly hallucinated response? So, is there any mechanism for retrying, a human escalation path or even a graceful error state?

5. User Stories & Use Cases

- **User Stories:** *Example:* "As a product manager, I want to enter a prompt using my own way of describing so that it provides me with a design without manual drawing by me."
- **Use Cases (Main Interactions):**

  Prompt to UI Flow: [e.g. Users enter the prompt, then Stitch parses the intent, identifies required components, and then renders a customizable 2D Canvas.]

  Flow Generation: [e.g. User selects a component, and triggers/enables the section called "Generate Next Screen". So, Stitch generates the target screen while maintaining the digital token i.e. color/fonts etc.]

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- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- [e.g. A conversational canvas for users to generate and edit in 2D workspace.]
- [e.g. Coherent and consistent design flow for entire system]
- [e.g. One click export feature along with codes.]
- [e.g. Support both text to UI & picture to UI feature]

7. Features Not Included (Scope Control)

- [e.g. Implementation of backend logic, API, & database]
- [e.g. Only for frontend scaffolding, does not support production environment]
- [e.g. Multi-user inputs simultaneously are not supported [Initial Versions]]

8. Assumptions & Constraints

- **LLM Cost Constraint:** State your team's estimation on token cost per average user session. Do identify at least one design related decision that is made specifically to keep the inference costs manageable at scale. [e.g. caching repeated queries].
- **Technical Constraints:** [e.g. Code export features are only limited to export in HTML, CSS. Other modern frameworks like React/ Swift (IOS) are not supported.]
- **Performance Constraints:** [e.g. To design high-quality visuals, it requires many tokens, so there is a generation limit.]
- **User Input:** [e.g. The model is unable to generate without human prompts and is not ready for production so requires human approval.]

9. Risks & Questions Throughout Development

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- **Design Similarity: [e.g.** How are we ensuring that AI is not generating similar designs across users?]
- **Frame Stability: [e.g.** Does the user need to fix the layout if it is found broken after exporting?]
- **Drawing Interpretation**: [e.g. What will happen if the AI is unable to contextualize the sketch that is provided?]

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